AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 3 · STUDENT LESSON

Sensors Are Messengers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 2: distinguish input, processing and output.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Describe a measurement using its quantity, value, unit and conditions. Try the recall before opening notes.

Your learning target

I can describe a **measurement** using its quantity, value, **unit** and conditions.

I can distinguish a **sensor** reading from an interpretation or decision.

I can use repeated comparisons to investigate a **measurement** problem without inventing certainty.

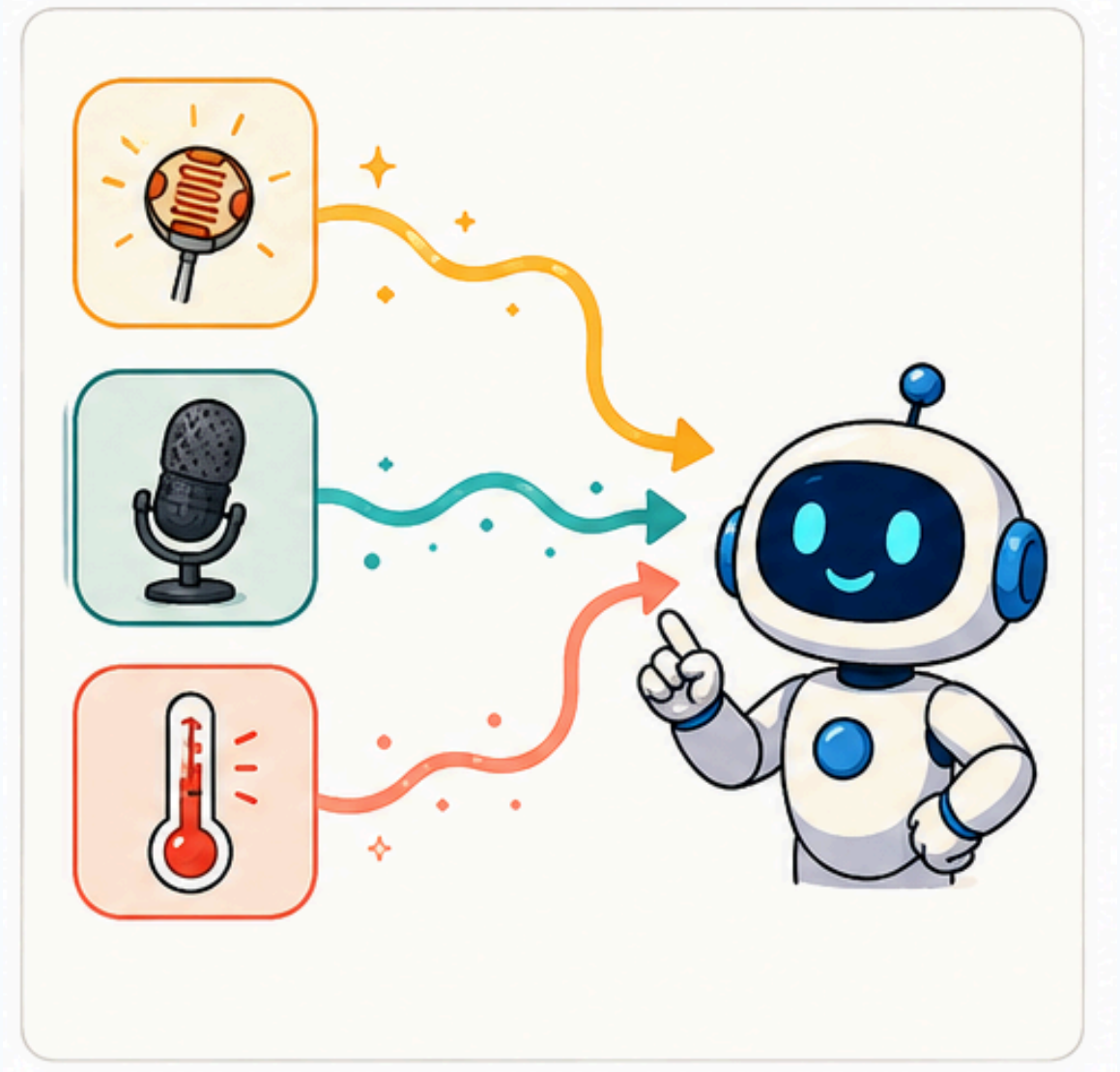
Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from **Input**, **Process**, **Output**. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Sensors measure **signal**s; later **process**ing gives those **signal**s meaning.



Your target

LEARNING TARGET Two light **sensor**s report 40 and 80 in different rooms. Their settings and **unit**s are unknown. Can you conclude the second room is twice as bright?

> Design a fair comparison without using a camera or personal **data**.

Learner name:

Readiness for the connected explanation

A record retains only the highest temperature of a day. Can it establish when that temperature occurred? Explain using two possible records.

This week's end goal

By the end, I can compare **sensor** readings while controlling where and when they are measured.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Sensors Are Messengers case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY A classroom weather station measures light, sound, and temperature but does not decide what those **measurement**s mean.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

A garden display shows 24. Is that warm? We cannot interpret the number until we know what was measured and the **unit**.

It might mean 24 degrees Celsius, 24 percent soil moisture according to a particular device, or 24 hours since watering. The screen alone does not supply the missing meaning.

A useful **measurement** record names the quantity, value, **unit**, time and relevant conditions. For air temperature, the location and exposure matter: a device in direct sunshine may warm differently from one in shade.

Record the conditions rather than assuming that every nearby device measured the same situation.

Measurement is a way of representing part of the physical world. It does not capture everything about that world.

One air-temperature reading does not tell us how dry the roots are, whether a plant is diseased or whether watering is permitted.

STUDENT LESSON · 3 OF 18

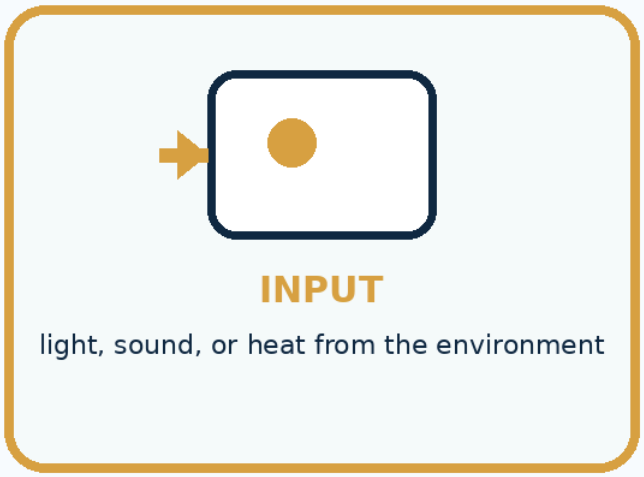
See how it works

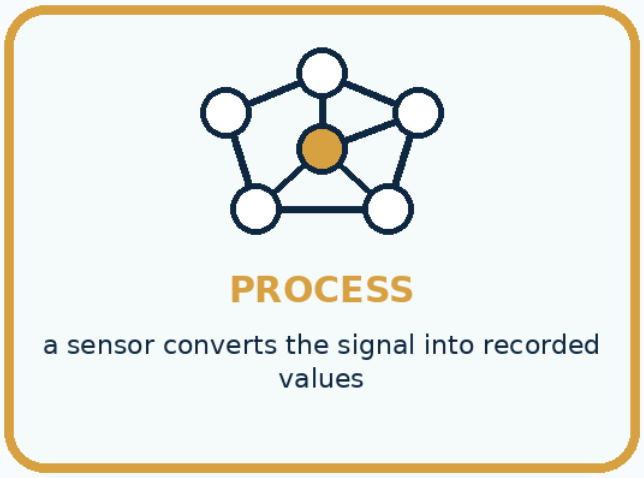
> Follow the mechanism for Sensors Are Messengers. Connect each labelled input, deciding step and result.

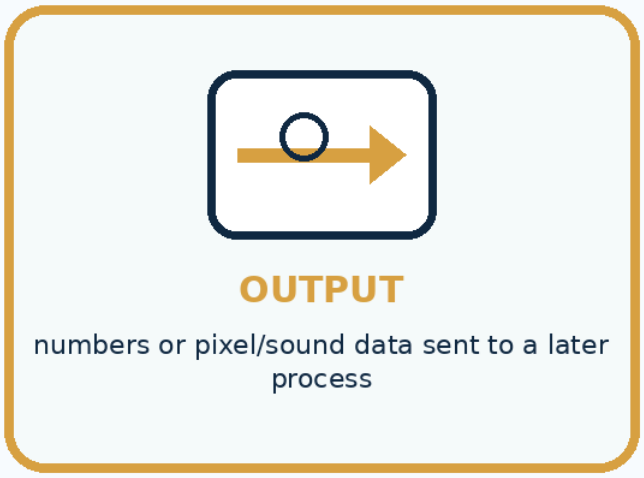
VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT light, sound, or heat from the environment

PROCESS a **sensor** converts the **signal** into recorded values

OUTPUT numbers or pixel/sound **data** sent to a later **process**

Key idea

CORE IDEA **Sensor**s measure **signal**s; later **process**ing gives those **signal**s meaning.

How the mechanism works

UNDER THE HOOD A **sensor** is a transducer: it converts a physical quantity into a **signal** a computing system can represent. Sensing and interpretation are separate.

A camera measures light; an image model may later infer a label.

Draw a fresh example with all three stages:

A closer explanation

Three different stages in a garden system

INPUT Physical temperature at the sensor	OPERATION Sensor and measuring electronics	OUTPUT Recorded air temperature 24 degrees C
INPUT Recorded value 24 degrees C	OPERATION Written rule checks whether value exceeds 28	OUTPUT Warning is off
INPUT Readings and a gardener's inspection	OPERATION Gardener decides what action is appropriate	OUTPUT An action or a decision to wait

Sensing, checking a rule and deciding what to do are separate stages. A correct rule can still receive an unsuitable measurement.

The **sensor** responds to a physical quantity. Electronics convert the response into a recorded value. A later program can compare the value with a threshold.

Nothing about this path requires us to claim that the **sensor** understands plant health.

The rule uses exceeds 28, so a reading of exactly 28 does not trigger it. At 29 it does.

This boundary is a property of the **written rule**; it is separate from the question of whether the measuring device is accurate enough for the intended purpose.

The sensor measures. The rule decides.

This model uses a made-up light scale, from 0 (darker) to 100 (brighter).

Light reading: 18

Rule: if reading is below 30, turn the lamp on; otherwise turn it off.

Lamp ON

Try 29, then 30. Which part measured the light, and which part chose the action?

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

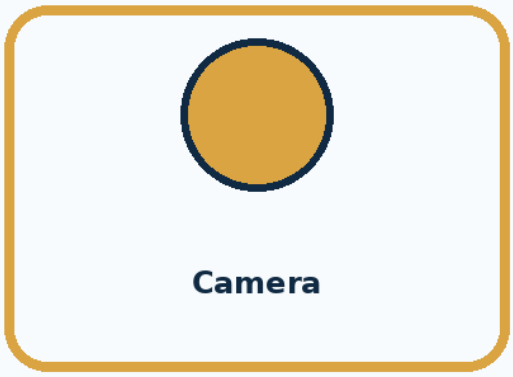
Compare the cases

> Compare the supplied cases for Sensors Are Messengers. Identify the detail that changes the conclusion.

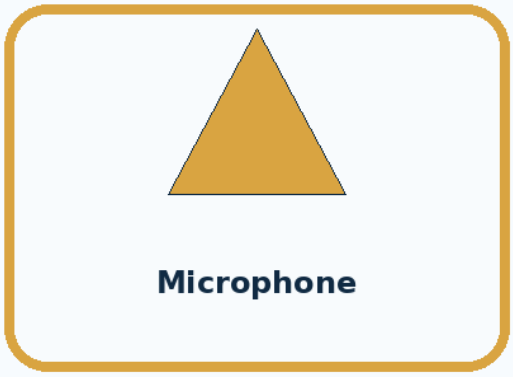
PICTURE GALLERY

Four Cases, Four Clues

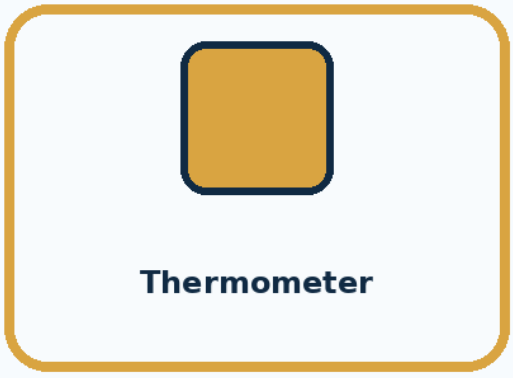
Use the visual cue, then read the **evidence** carefully.



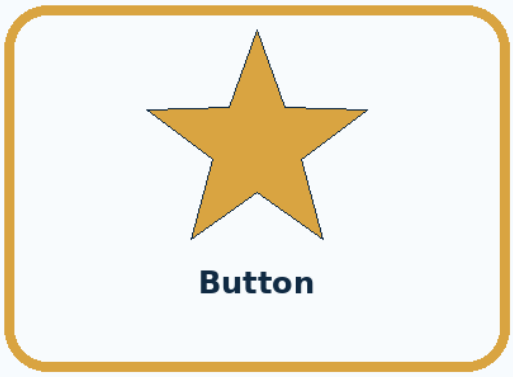
Camera



Microphone



Thermometer



The case to investigate

CASE 1 — Camera Clue: light becomes pixel values. Meaning: **measurement**, not meaning.

The case to investigate

CASE 2 — Microphone Clue: air-pressure changes become samples. Meaning: **measurement**, not understanding.

The case to investigate

CASE 3 — Thermometer Clue: heat becomes a temperature number. Meaning: **measurement**, not a weather decision.

The case to investigate

CASE 4 — Button Clue: a press becomes an on/off value. Meaning: a human event becomes **data**.

Keep this boundary clear

BOUNDARY Do not say a **sensor** sees, hears, or understands. It measures and sends a representation.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Move only B into brighter light.

Predicted result

Prediction: The next gap could change for reasons other than a difference in device accuracy.

Why the result follows

Reason: The comparison now changes location as well as device; there is no supplied reading from the new location.

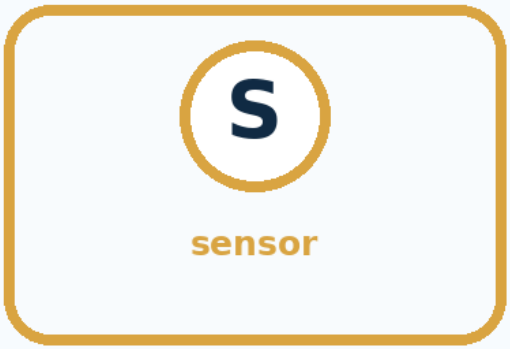
Words that unlock the idea

> Use the vocabulary of Sensors Are Messengers to explain a concrete decision, rather than only memorising definitions.

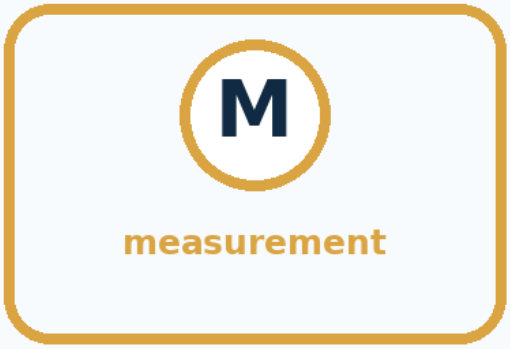
VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
sensor	device that measures a physical signal	
signal	a changing quantity that can be measured	
measurement	a recorded value describing part of a signal	
data	represented information available for processing	

Misconception detective

Find the unsupported claim

CLAIM TO REPAIR The camera knows there is a cat.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Unit	An agreed amount used to express a quantity, such as a second or a degree Celsius
Variation	Differences among readings or observations
Calibration	Checking and, when appropriate, adjusting a measuring system against a reference

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STUDENT LESSON · 6 OF 18

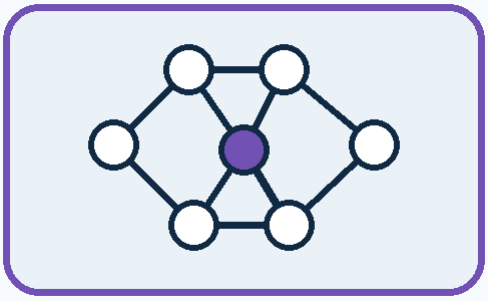
Follow a worked example

> Follow the worked trace for Sensors Are Messengers, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE A classroom weather station measures light, sound, and temperature but does not decide what those **measurement**s mean.

Step 1

STEP 1 — OBSERVE light, sound, or heat from the environment

Step 2

STEP 2 — TRACE a **sensor** converts the **signal** into recorded values

Step 3

STEP 3 — CONCLUDE numbers or pixel/sound **data** sent to a later **process**

Model answer

MODEL REASONING 820 could be a light **measurement**, 24 C comes from a temperature **sensor**, and changing sound values come from a microphone.

The numbers alone do not prove whether the plant needs water or whether a voice was present; later **process**ing must interpret them.

A claim to avoid

DO NOT OVERCLAIM Do not say a **sensor** sees, hears, or understands. It measures and sends a representation.

What new **evidence** could change this answer?

A closer explanation

Worked explanation · Sensors Are Messengers

Two devices are placed side by side in a shaded room and allowed to settle. At three agreed times, device A reads 20, 21 and 20 degrees C; device B reads 22, 23 and 22 degrees C. B is 2 degrees higher on every comparison.

That pattern is **evidence** of a consistent difference under these test conditions.

Time	A in degrees C	B in degrees C	B minus A
10:00	20	22	2
10:05	21	23	2
10:10	20	22	2

Worked explanation · Sensors Are Messengers

We cannot conclude from agreement patterns alone that A is correct and B is wrong. A suitable checked reference can help investigate.

We also need to consider whether the devices have different response times, resolutions or placement. Three comparisons in one room do not prove performance in every temperature range.

Calibration is a reference check, sometimes followed by an adjustment. Do not simply subtract 2 from every future B reading because these three pairs happened to differ by 2.

A correction needs support over the conditions where it will be used.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sensors Are Messengers

Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken.

The facts we will use

Supplied facts

On the same shelf, simultaneous readings are A: 40,50,60 and B: 45,55,65 in declared toy light **unit**s.

Supplied facts

Neither device has been compared with a trusted reference.

AI ACADEMY / AI-1/3

Simultaneous light readings

	Sensor A	Sensor B	B minus A
Time 1	40	45	5
Time 2	50	55	5
Time 3	60	65	5

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/3

Sensors Are Messengers

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.



1. Same shelf

Place both devices together and allow them to settle.

↓



2. Paired readings

Record three matching times: (40,45), (50,55), (60,65).

↓



3. Compare gaps

B minus A is 5, 5 and 5 units.

↓



4. Check a reference

A trusted reference is needed to decide which reading is closer.

CHANGE ONE THING
Move only B into brighter light.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Same shelf	Place both devices together and allow them to settle.
2. Paired readings	Record three matching times: (40,45), (50,55), (60,65).
3. Compare gaps	B minus A is 5, 5 and 5 units.
4. Check a reference	A trusted reference is needed to decide which reading is closer.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A repeated difference of 5 supports investigating an offset between devices. Matching time and location removes two obvious comparison problems. It does not show which device is correct: both could have errors.

Preserve paired readings instead of adjusting them silently.

What this example does not establish

Keep the stated limit

Three pairs do not prove an offset holds at every light level, nor identify the accurate device.

Keep the stated limit

Read the labelled picture: Why can we calculate the earlier gap but not predict an exact new B reading?

Keep the stated limit

Change one thing: Move only B into brighter light. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Sensors Are Messengers claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Camera	light becomes pixel values	measurement, not meaning

Case	Evidence	What it shows
Microphone	air-pressure changes become samples	measurement, not understanding
Thermometer	heat becomes a temperature number	measurement, not a weather decision
Button	a press becomes an on/off value	a human event becomes data

Repair two tempting claims

Challenge claim A

CLAIM A The camera knows there is a cat.

Repair:

Challenge claim B

CLAIM B More **sensor**s always make a system better.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

A warning based on one temperature reading can mean the recorded value crossed a threshold. It cannot automatically mean a plant needs water. That broader claim would need different **evidence**.

When a system makes a surprising decision, check the **unit**s, time, location, **measurement** and rule in order.

Check the claim against the facts

For classroom investigations, use teacher-provided records or ordinary safe room conditions. The reasoning task does not require heating devices, opening electronics or testing people.

A clear paper investigation can reveal the same distinction between a reading and an interpretation.

A display says 30 without a unit. State two questions needed before using it in a temperature-warning rule

STUDENT LESSON · 8 OF 18

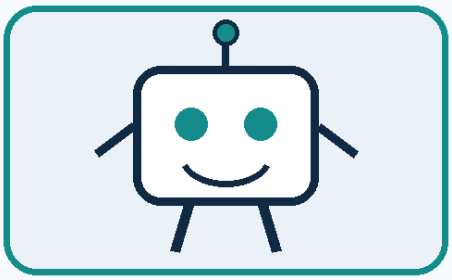
Practise together

Practise Sensors Are Messengers with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: **Sensor**s measure **signal**s; later **process**ing gives those **signal**s meaning.

Use a fresh example to explain **sensor**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: More **sensor**s always make a system better.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Suppose one classroom device reports 27 degrees C near a sunny window and another reports 22 degrees C in shade. The five-degree difference may reflect the locations. Calling one device faulty skips an alternative explanation.

First write a **prediction**: if location explains the difference, placing both devices together after they settle should reduce it.

A fair comparison changes the suspected factor while keeping other relevant conditions as similar as practical. Record what was changed and what remained uncertain.

Repeated results can strengthen or weaken an explanation, but a short classroom test is not a guarantee about all future **measurement**s.

> **Two devices disagree near a window and a shaded wall. Describe a comparison that helps separate location from device differences**

Practise with less help: Trace a boundary

Use workbook W2 for the connected example "**Sensor**s and **measurement**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Sensor**s and **measurement**. Proposed explanation: "The more stable device must be accurate". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Why can we calculate the earlier gap but not predict an exact new B reading?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

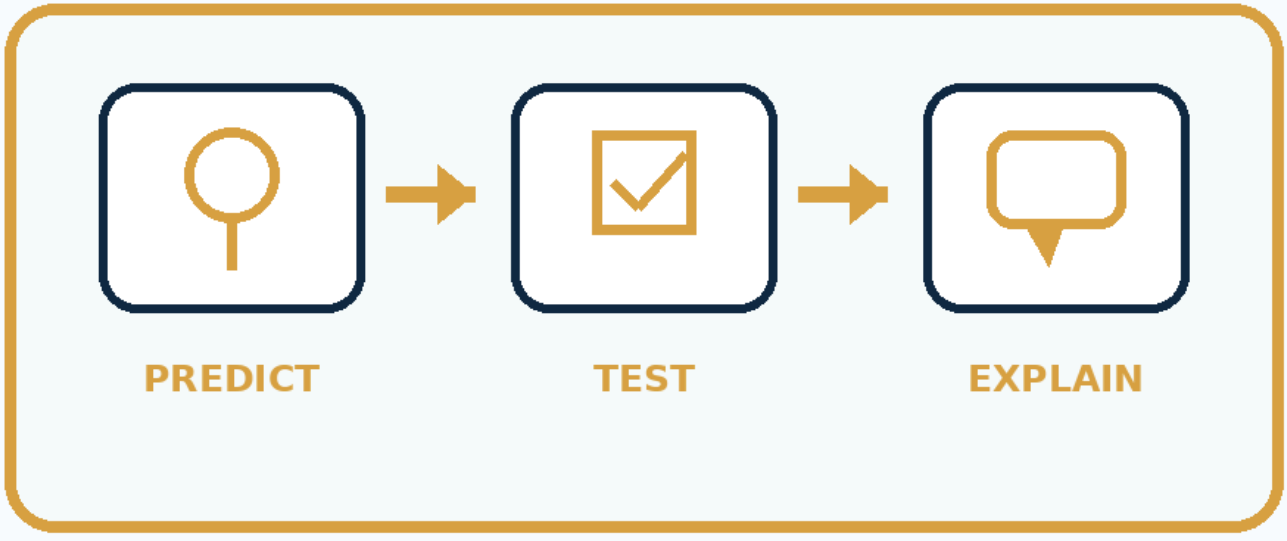
Run the investigation

> Investigate Sensors Are Messengers. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Signal Messenger Match

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Match camera, microphone, and temperature **sensor**s to **data**.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

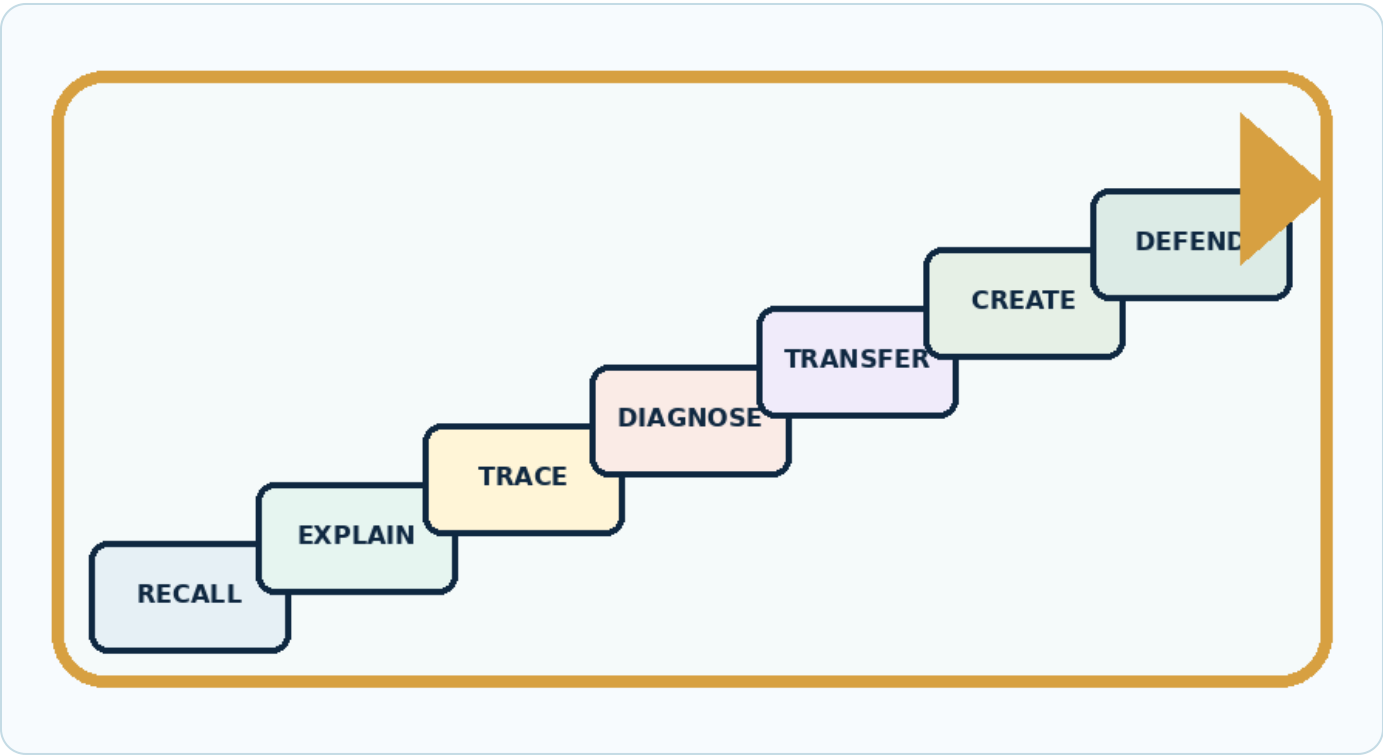
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Sensors Are Messengers. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **sensor**.

EXPLAIN **Sensor**s measure **signal**s; later **process**ing gives those **signal**s meaning.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: The camera knows there is a cat.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

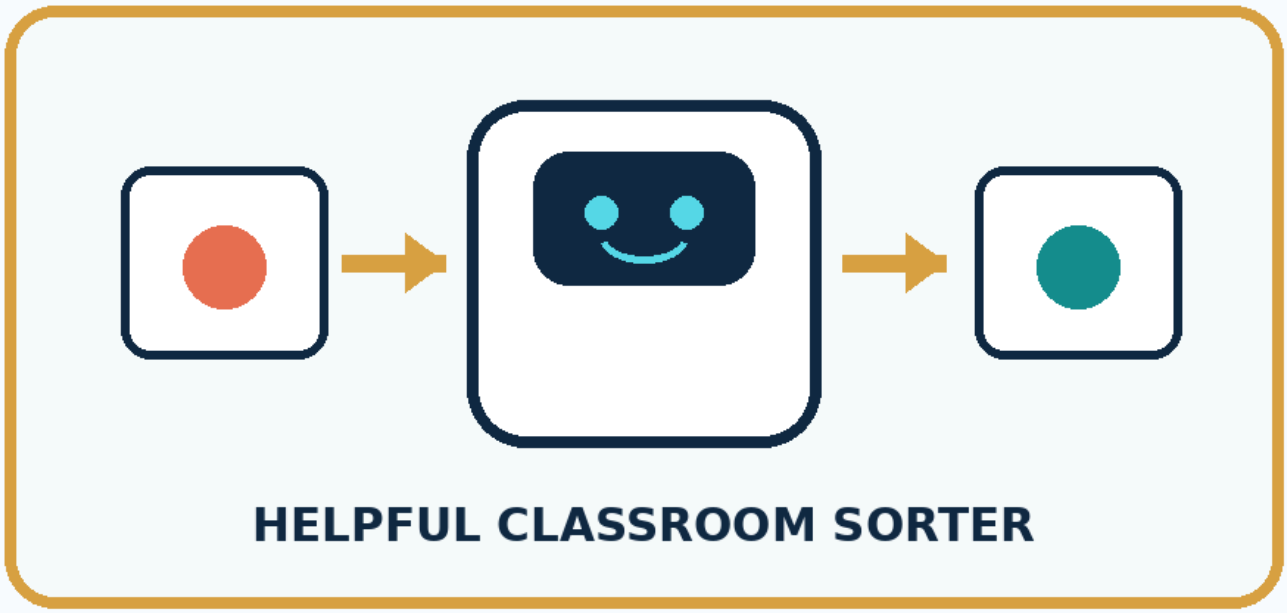
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Mark which **sensor**, if any, would collect each Classroom Sorter **input**; prefer direct human entry when a **sensor** is unnecessary.

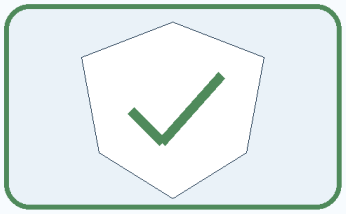
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN Do not say a **sensor** sees, hears, or understands. It measures and sends a representation.

My project decision:

Evidence for your project

Save sorter **measurement**s with device, location and time so unusual **input**s can be checked before decisions.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF A plant station reports 820, 24 C, and a sequence of sound values.

Identify each likely **sensor**, its **signal**, and what remains unknown.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK 820 could be a light **measurement**, 24 C comes from a temperature **sensor**, and changing sound values come from a microphone.

The numbers alone do not prove whether the plant needs water or whether a voice was present; later **process**ing must interpret them.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Rules or Learned Patterns?.

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Sensors Are Messengers case for supported practice. Repeated

> examples do not become fresh assessments.

Week 3 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A light **sensor** reads 15 in a cupboard and 90 near a window. A microphone records a clap.

The question

> Which measures a light-related **signal**? Can either directly measure happiness?

Worked explanation

The light **sensor** measures light; the microphone measures sound. Neither directly measures feelings. **Unit**s and **calibration** matter when comparing devices.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Two light readings, one sensor

Evidence or stage	Value or result
Cupboard	15
Window	90

A light **sensor** reads 15 in a cupboard and 90 near a window. A microphone records a clap.

> Which measures a light-related **signal**? Can either directly measure happiness?

The light **sensor** measures light; the microphone measures sound. Neither directly measures feelings. **Unit**s and **calibration** matter when comparing devices.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Investigate a new measurement dispute

A classroom sound display near a speaker reads higher than one near the door. A student declares the door device broken. Propose a safe comparison using normal classroom sound.

Predict one result consistent with location explaining the difference and one result that would leave a device difference worth investigating.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 3 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A station reports light level 18 and temperature 22 degrees C. Which **sensor** produced each reading? Can either reading alone tell whether a child feels comfortable? Explain.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Mark which **sensor**, if any, would collect each Classroom Sorter **input**; prefer direct human entry when a **sensor** is unnecessary.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Two light **sensor**s report 40 and 80 in different rooms. Their settings and **unit**s are unknown. Can you conclude the second room is twice as bright? Design a fair comparison without using a camera or personal **data**.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate Sensors Are Messengers, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken.

By the end, I can	How I will show it
Describe a measurement using its quantity, value, unit and conditions	Explain the deciding step, show a trace or test, and state one limit.
Distinguish a sensor reading from an interpretation or decision	Explain the deciding step, show a trace or test, and state one limit.
Use repeated comparisons to investigate a measurement problem without inventing certainty	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Sensors Are Messengers without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Describe a **measurement** using its quantity, value, **unit** and conditions

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? A trusted reference is needed to decide which reading is closer.

1. Compare gaps

2. Check a reference

3. Paired readings

4. Same shelf

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check a reference A trusted reference is needed to decide which reading is closer. A repeated difference of 5 supports investigating an offset between devices.

Matching time and location removes two obvious comparison problems. It does not show which device is correct: both could have errors. Preserve paired readings instead of adjusting them silently.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move only B into brighter light. Which conclusion is supported by the supplied facts?

1. The next gap could change for reasons other than a difference in device accuracy.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The next gap could change for reasons other than a difference in device accuracy.

> The comparison now changes location as well as device; there is no supplied reading from the new location.

Repair a missing building block

Review **Input**, **Process**, **Output** (ai-1/2). Goal: Trace information through a system without confusing information with a physical action. Short practice: Use the richer record to find blue pencils in Box B.

Explain why total alone was insufficient.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

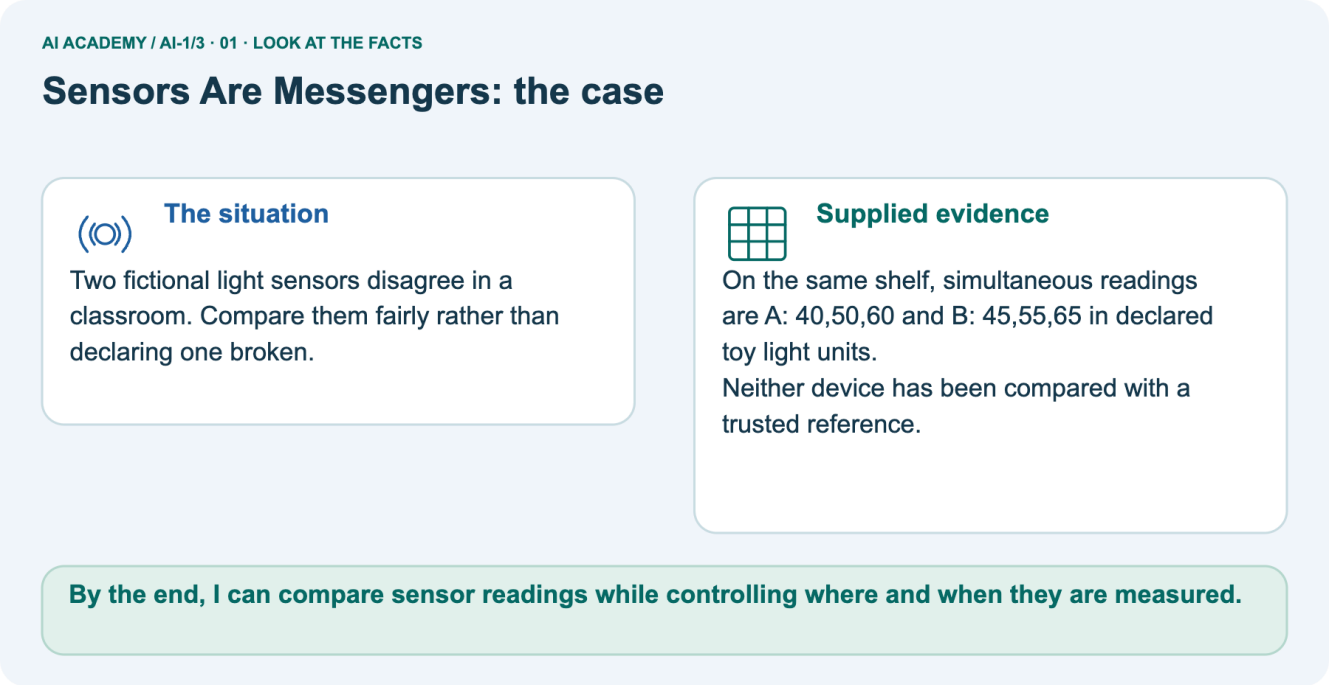
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Sensors Are Messengers graphic using its labels, deciding evidence and relevant limitation.

01. Sensors Are Messengers: the case



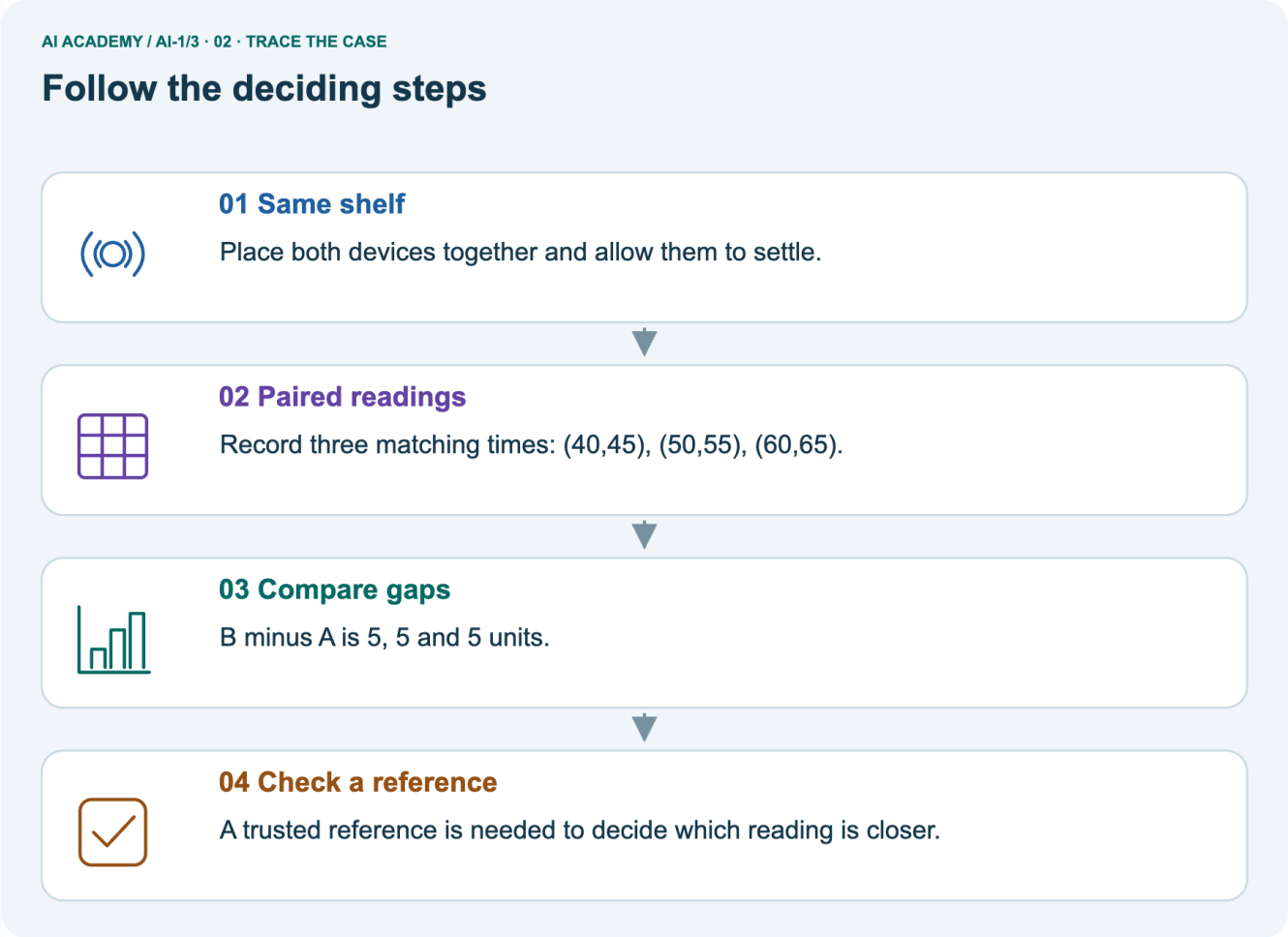
Follow the labelled picture with the matching explanation. Open the vector diagram

Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken. Supplied facts: On the same shelf, simultaneous readings are A: 40,50,60 and B: 45,55,65 in declared toy light **unit**s.

Neither device has been compared with a trusted reference.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

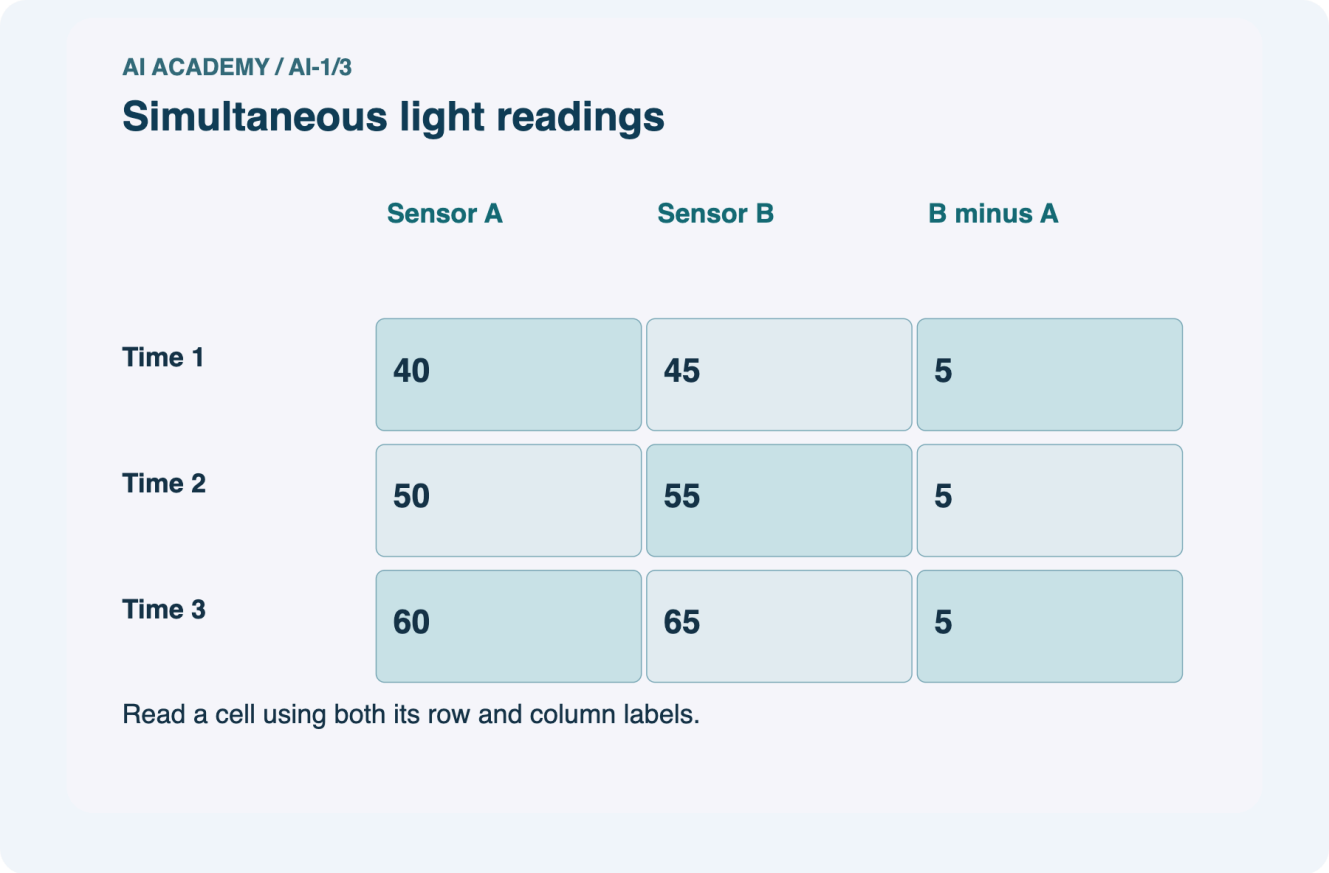


Follow the labelled picture with the matching explanation. Open the vector diagram

Same shelf: Place both devices together and allow them to settle. → Paired readings: Record three matching times: (40,45), (50,55), (60,65). → Compare gaps: B minus A is 5, 5 and 5 **unit**s. → Check a reference: A trusted reference is needed to decide which reading is closer.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Simultaneous light readings

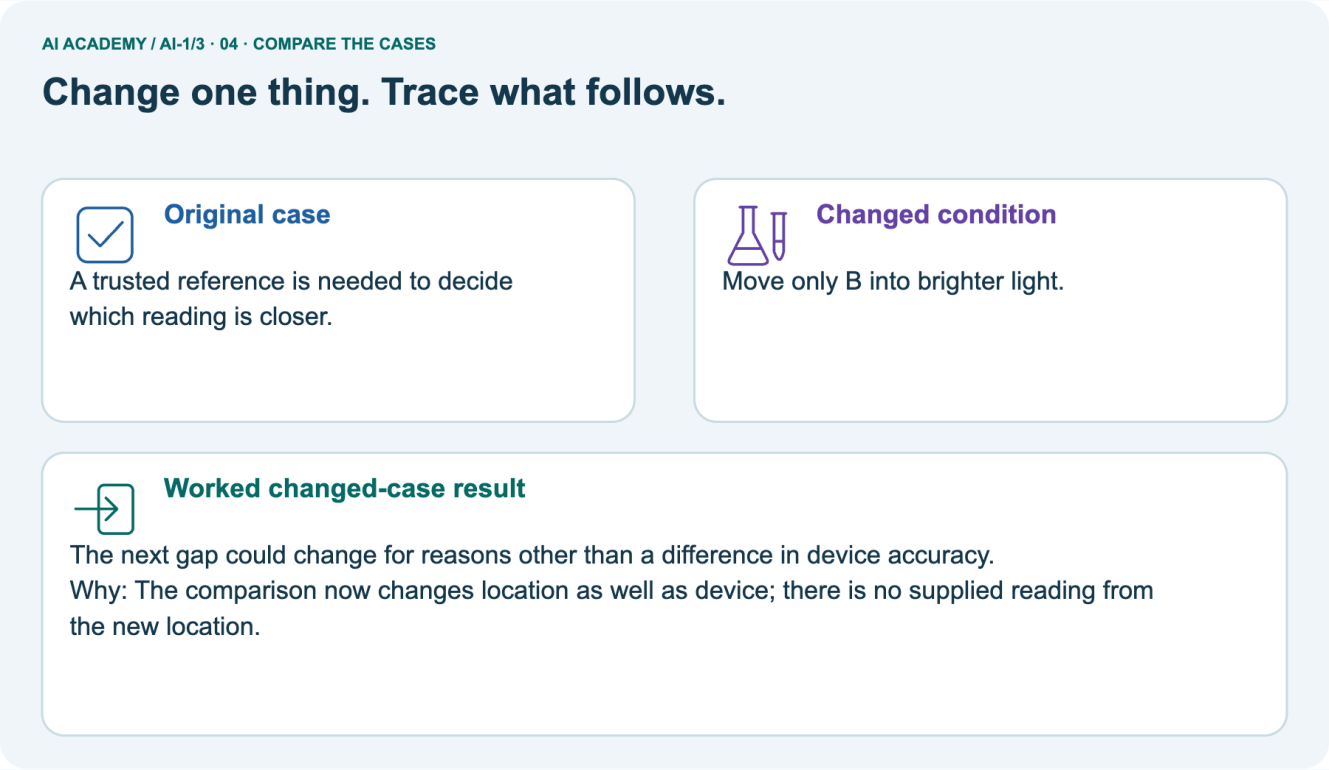


Follow the labelled picture with the matching explanation. Open the vector diagram

Simultaneous light readings. Time 1: **Sensor** A = 40, **Sensor** B = 45, B minus A = 5. Time 2: **Sensor** A = 50, **Sensor** B = 55, B minus A = 5. Time 3: **Sensor** A = 60, **Sensor** B = 65, B minus A = 5.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



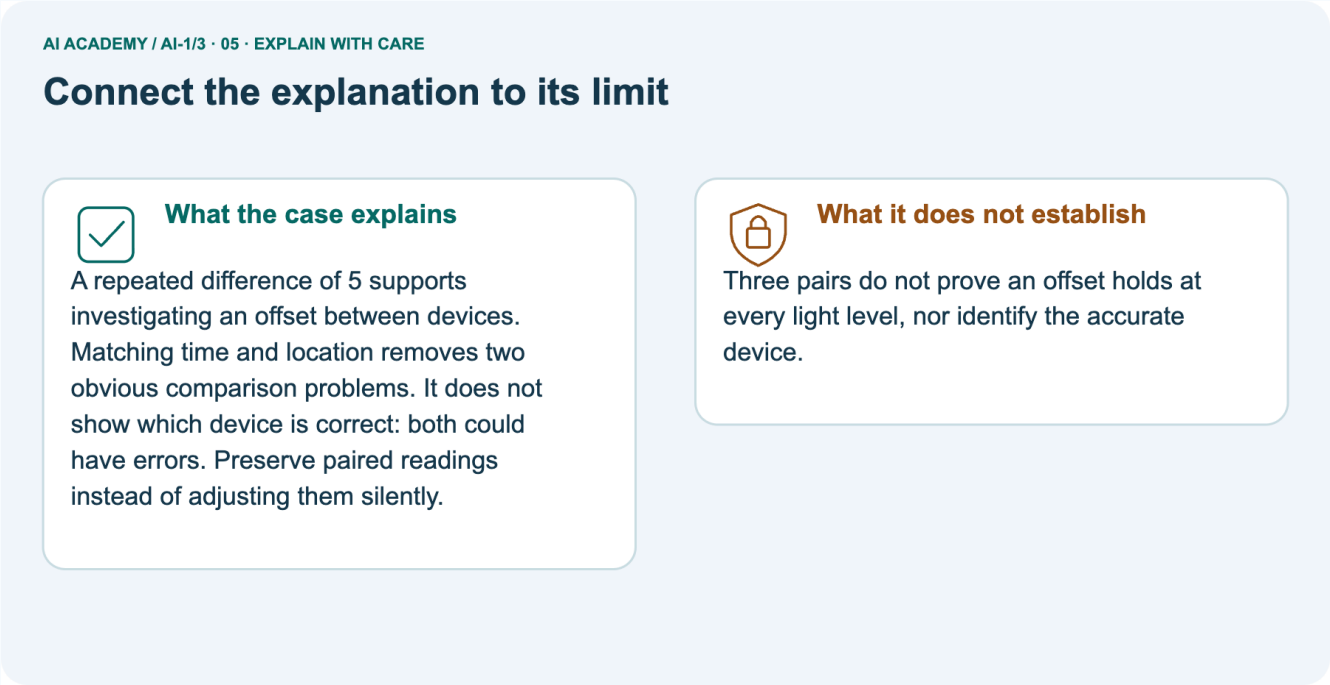
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A trusted reference is needed to decide which reading is closer. Changed condition: Move only B into brighter light.

Expected result: The next gap could change for reasons other than a difference in device accuracy. Reason: The comparison now changes location as well as device; there is no supplied reading from the new location.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



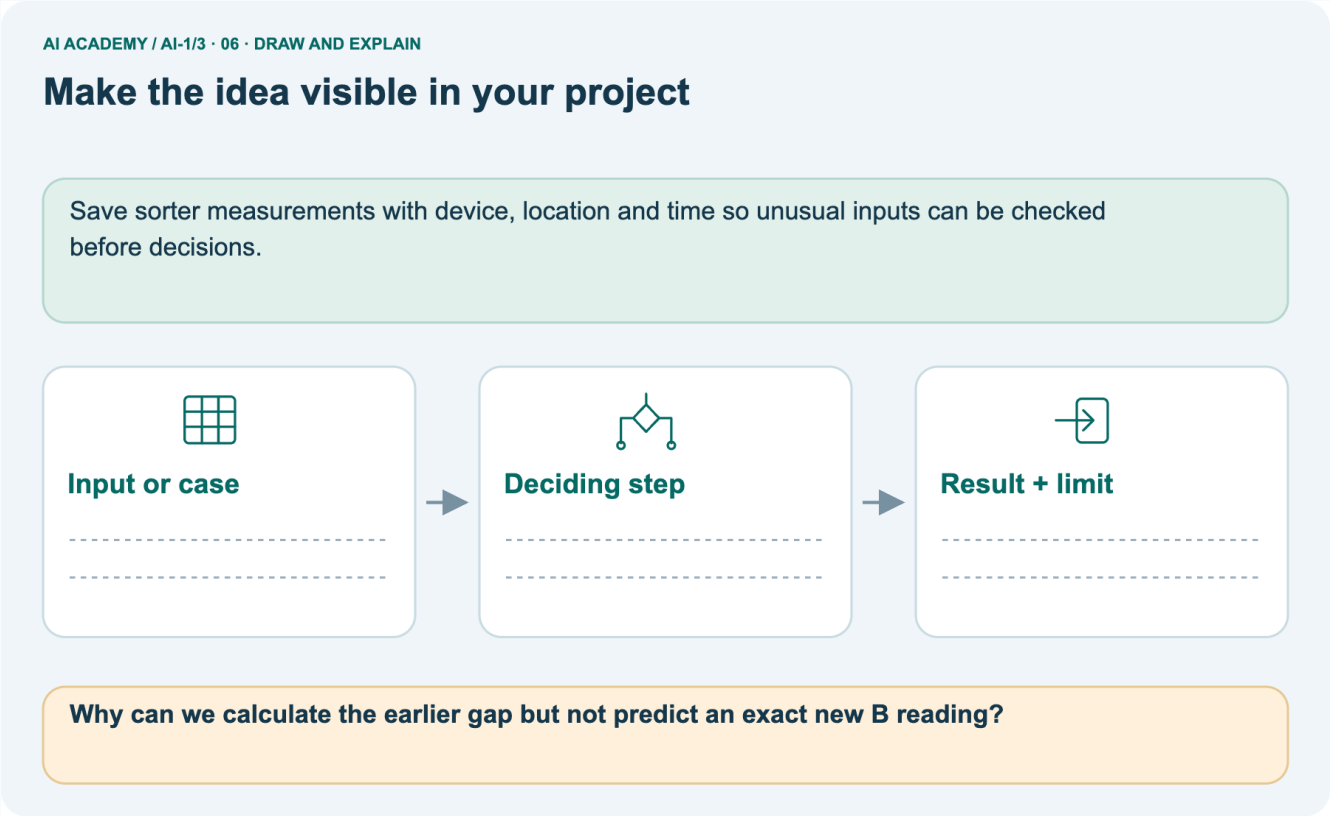
Follow the labelled picture with the matching explanation. Open the vector diagram

A repeated difference of 5 supports investigating an offset between devices. Matching time and location removes two obvious comparison problems. It does not show which device is correct: both could have errors.

Preserve paired readings instead of adjusting them silently. Limit: Three pairs do not prove an offset holds at every light level, nor identify the accurate device.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save sorter **measurement**s with device, location and time so unusual **input**s can be checked before decisions. Draw your **input**, deciding step, result and limit.

Supported question: Why can we calculate the earlier gap but not predict an exact new B reading?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.